HAMED HOJATIAN

Ph.D. Candidate

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EDUCATION

Ph.D. in Applied Deep Learning in wireless communication Systems

Polytechnique Montréal

Fall 2019 - current — Montreal, Canada

Supervisor: **Prof. François Leduc-Primeau**
Co-Supervisor: **Prof. Jean-François Frigon**
CGPA: **4.0/4.0**

M.Sc. in Telecommunication Engineering

Isfahan University of Technology (IUT)

2014 - 2017 — Isfahan, Iran

Supervisor: **Prof. Mohammad Javad Omidi**
CGPA: **3.5/4.0**

B.Sc. in Electrical Engineering

Babol (Noshirvani) University of technology (NIT)

2010 - 2014 — Mazandaran, Iran

Supervisor: **Prof. Hosein Miar-Naeimi**
CGPA: **3.1/4.0**

DISSERTATION WORK

Ph.D. Thesis

Hybrid Beamforming in Massive MIMO Systems with Deep Learning

- RSSI-Based hybrid beamforming design with deep learning;
- Improve the spectral and energy efficiency of hybrid beamforming design;
- Unsupervised learning for hybrid beamforming design;
- Coordinated hybrid beamforming design with deep neural network; and
- ...

M.Sc. Thesis

Time and Frequency synchronization in Massive MIMO systems

- Carrier Frequency Offset (CFO) and channel estimation Using ZC sequence;
- Detection in receiver using iterative method; and
- Improve the method using Filtered- OFDM.

B.Sc. Thesis

OFDM Transmitter and receiver

- Developed in MATLAB simulation
- OFDMA multi-user simulation in AWGN channel.

SEEKING FOR

"To work in a progressive and dynamic research organization where one could solve challenging scientific problems"

PROG. SKILLS

Python	●●●●○
Pytorch	●●●●●
JAVA SCRIPT	●●●○○
MATLAB	●●●●○
LaTeX	●●●●○
MS Office	●●●●○
HTML5	●●●○○
CSS	●●●○○
Visual basic	●●○○○
C / C++	●●○○○
Bash	●●○○○

LANGUAGES

English	●●●●○
French	●○○○○
Persian	●●●●●

STRENGTHS

Hard-working
Decision Making
Ability to Work Under Pressure
Motivator & Leader
Conflict Resolution
Time Management
Target oriented
Flexibility and Adaptability

TELECOMM. TOOLS

Ericsson OSS	●●●●○
MapInfo	●●●●○
Business Object (BO)	●●●○○
Atoll	●●●○○
TEMS	●●○○○
Schema	●●○○○
Actix	●●○○○

PUBLICATION

- H. Hojatian, J. Nadal, J.-F. Frigon, and F. Leduc-Primeau. **"Unsupervised Deep Learning for Massive MIMO Hybrid Beamforming"**, Submitted to IEEE Transactions on Wireless Communications, arXiv:2007.00038. (2nd review round)
- H. Hojatian, J. Nadal, J.-F. Frigon, and F. Leduc-Primeau. **"Two Unsupervised Deep Learning Approaches forCoordinated Hybrid Beamforming"**, submitted to 2021 IEEE International Conference on Communications (ICC): Wireless Communications Symposium (IEEE ICC'20 - WC Symposium), 2021.
- H. Hojatian, V. N. Ha, J. Nadal, J.-F. Frigon, and F. Leduc-Primeau. **"RSSI-Based Hybrid Beamforming Design with Deep Learning"**, 2020 IEEE International Conference on Communications (ICC): Wireless Communications Symposium (IEEE ICC'20 - WC Symposium), Dublin, Ireland, 2020.
- H. Hojatian, M. J. Omidi, H. Saeedi-Sourk, A. Farhang, **"Joint CFO and channel estimation in OFDM-based massive MIMO systems"**, 2016 8th International Symposium on Telecommunications (IST), Tehran, 2016, pp. 343-348, doi: 10.1109/ISTEL.2016.7881837.

RESEARCH INTERESTS

- Applied ML in telecommunication;
- Deep Learning;
- Reinforcement learning;
- Signal processing in MASSIVE MIMO communication;
- Millimeter wave in Massive MIMO and Antenna array;
- Coordinated, cooperative beamforming and distributed antenna; and
- Cell free communication.

WORK EXPERIENCE

Senior Telecommunication Engineer

Mobile Telecommunication Company of Iran (MCI)

Nov 2015 - Nov 2018 | Tehran, Iran

- Ericsson OSS in Tehran Network;
- Network Planning (2G, 3G and LTE);
- Network Optimization (BSC owner and RNC owner);
- Project manager of RNC rehoming project in Tehran; and
- Parameter tuning.

Vice-President

Eleventh (11th) national robotics competition at BABOL University of Technology

Oct 2015 | Mazandaran, Iran

COURSES

Polytechnique Montréal (PolyMTL)

- Learning Representation (A+) - instructor: Prof. Aaron Courville
- Wireless Communication (A) - instructor: Prof. Jean-François Frigon
- Estimation and Detection Theory (A) - instructor: Prof. Jean-François Frigon

Isfahan University of Technology (IUT)

- Advance Digital Communication (ADC)
- Information Theory
- Software Define Radio (SDR)
- Digital Signal Processing (DSP)

TALKS & TEACHING

Author

IEEE ICC2020

Jun 2020 | Virtual-Dublin, Ireland

- Present my accepted paper in ICC2020;

Teaching assistant

BNUT

2013 | Mazandaran, Iran

- Teaching assistant for Electronic I
- Teaching assistant for Electronic circuit I
- Teaching assistant for Digital Signal Processing

REFEREES

Dr. François Leduc-Primeau
@ francois.leduc-primeau@polymtl.ca
Assistant Professor, Department of Electrical Engineering, Polytechnique Montréal, Canada

Dr. Jean-François Frigon
@ jf.frigon@polymtl.ca
Full Professor, Department of Electrical Engineering, Polytechnique Montréal, Canada

Dr. Mohammad Javad Omidi
@ omidi@cc.iut.ac.ir
Assistant Professor, Department of Electrical Engineering, Isfahan University of Technology, Iran

HOBBIES

Music
Listen quality music of every genre

Searching
Always eager to find the updated topics in other fields

Swimming
Fresh water is my charger

Travelling
Always fond of traveling to exotic places

CERTIFICATES

- "**4TH IVADO/MILA SCHOOL IN DEEP LEARNING**" *IVADO/MILA*, 2019, Montreal, Canada.
- "**DEEP LEARNING WITH PYTORCH**" *Udemy*, 2018, Online.
- "**LTE OPTIMIZATION OF RADIO NETWROK (3GPP)**" *TECHCOM Consulting GmbH*, 2018, Tehran, Iran.
- "**LTE COUNTER AND KPI (ERICSSON)**" *TECHCOM Consulting GmbH*, 2017, Tehran, Iran.
- "**LTE RADIO NETWORK PLANNING**" *TECHCOM Consulting GmbH*, 2017, Tehran, Iran.